

Texts, Science and the Performing Arts - The Foundation Track

Chapter F5.6 — From Zero: Languages and the Classical Literature, Ancient Indian Science and the Universities, the Eight Classical Dances, the Two Music Systems, and the Painting Schools

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NOTE · HOW TO USE THIS DOCUMENT

The chapter that completes the cultural half of Module 5, and it is built almost entirely of pairings - a language with its status, a scientist with his work, a dance with its state, a painting school with its patron. Session 1 takes languages and the classical literature, including the official list of classical languages, which is recent enough that most guidebooks print it wrongly. Session 2 takes ancient Indian science, where the honest position matters: some claims are solid and demonstrable, others are inflated, and the chapter separates them. Session 3 takes the eight classical dance forms with their states and the Natyashastra behind them. Session 4 takes the two systems of classical music and the painting schools from Mughal miniatures to the folk traditions, which were held back from F5.5 because they belong with the courts and communities that produced them. Study time 3 hours.

This is the FOUNDATION layer. The list of official classical languages was expanded in October 2024 and now stands at ELEVEN; older sources print six, and this chapter gives the current position with the addition dates. Attributions of ancient scientific and medical texts to named individuals follow tradition, and several are compilations rather than single-author works. Where a claim about ancient Indian science is contested or has been overstated in popular writing, that is said plainly rather than repeated. Dates for the founders of the classical dance and music traditions are conventional.

Table of Contents

0. Why This Chapter Is Almost Entirely Pairings	3
1. Session 1 - Languages and the Classical Literature ★★★★★☆	4
1.1 The classical languages - the current list	4
2. Session 2 - Ancient Indian Science, Honestly ★★★★★☆	6
2.1 The ancient universities	8
3. Session 3 - Dance ★★★★★	8
3.1 The eight classical dances	9
4. Session 4 - Music and Painting ★★★★★	10
4.1 The painting schools	11
5. Graded Practice - 42 Questions ★★★★★	13
5.1 Level 1 - Recognition	13
5.2 Level 2 - Understanding	14
5.3 Level 3 - Recruitment Test standard	15
6. Answers and Explanations	17
6.1 Level 1 - answers	17
6.2 Level 2 - answers	18
6.3 Level 3 - answers	18
7. One-Minute Revision	20
8. Five-Minute Wall Sheet	22

0. Why This Chapter Is Almost Entirely Pairings

A language with its status, a scientist with his work, a dance with its state, a school with its patron. Learn the pairs and the chapter is done.

WHY THIS EXISTS · WHAT THIS CHAPTER IS, AND THE ONE HONESTY PROBLEM IN IT

This chapter completes the culture half of Module 5. **Structurally it is the simplest chapter in the module**, because almost every fact in it is a **two-column pairing**, and the paper asks for exactly that.

- 1 **Language to status** - which are the classical languages, and when each was recognised
- 2 **Scientist to work** - Aryabhata to the Aryabhatiya, Susruta to surgery, Nagarjuna to chemistry
- 3 **Dance to state** - eight forms, eight answers
- 4 **School to patron** - Mughal, Rajput, Pahari, Deccani; and the folk traditions to their regions

There is one place where this material needs care rather than memorisation, and it is Session 2. Ancient Indian science is a subject where **popular writing routinely overstates the evidence** - claims about aircraft, stem cells and plastic surgery in antiquity circulate widely and are not supportable. **The genuine achievements are substantial enough not to need inflating**, and this chapter states which is which. **That distinction is worth holding for its own sake, and it protects you in an interview**, where an overstated claim is the fastest way to lose a panel's confidence.

PYQ CONNECTION · WHERE THE MARKS SIT, AND ONE LIVE-DATA WARNING

- 1 **HIGHEST yield - the eight classical dances with their states.** A near-certain question type, and eight facts long
- 2 **HIGH yield - the classical languages.** And note the warning below, because this is the one topic in Module 5 where **the correct answer changed recently**
- 3 **HIGH yield - Aryabhata and the mathematics of zero and place value**, which also serves General Science
- 4 **MODERATE - the two music systems**, the **Carnatic trinity**, and the instrument classification
- 5 **MODERATE - the painting schools**, especially **Madhubani, Warli, Pattachitra and Kalamkari**, which appear in general-awareness questions as often as in history ones

THE LIVE-DATA WARNING. The official list of **classical languages was expanded in October 2024 from six to ELEVEN**. **Most printed guides, and a great many online compilations, still say six.** If a question offers **six** and **eleven**, the current correct answer is **eleven** - but be alert to the possibility of a paper written against the older list. **Section 1 gives both the old six and the five additions with their dates**, so you can answer either way and know which is which.

STUDY SESSION · CHAPTER F5.6 - THE FOUR SESSIONS

- 1 **Session 1 - Languages and literature:** the language families, Sanskrit, Prakrit and Pali, the classical-language list, and the literature not already covered (Section 1)
- 2 **Session 2 - Ancient Indian science, honestly:** mathematics, astronomy, medicine, metallurgy, the universities, and what is overstated (Section 2)
- 3 **Session 3 - Dance:** the Natyashastra, rasa, the eight classical forms with their states, and the major folk forms (Section 3)
- 4 **Session 4 - Music and painting:** Hindustani against Carnatic, the instrument classes, then the miniature and folk painting schools (Section 4)

Then **42 graded questions**.

1. Session 1 - Languages and the Classical Literature ★★★★★

Two great language families, three ancient literary languages, and a list of classical languages that changed in 2024.

START FROM ZERO · THE TWO FAMILIES, AND WHY THE DISTINCTION MATTERS

India's languages belong overwhelmingly to two families, and **the division is geographical as well as linguistic**.

- 1 **INDO-ARYAN** - the northern group, descended from the language of the Vedas. **Sanskrit, Prakrit, Pali**, and from them **Hindi, Bengali, Marathi, Gujarati, Punjabi, Odia, Assamese, Kashmiri, Sindhi, Nepali, Konkani** and others. **About three-quarters of Indians speak an Indo-Aryan language**
- 2 **DRAVIDIAN** - the southern group, and **not descended from Sanskrit**. **Tamil, Telugu, Kannada, Malayalam**, plus **Tulu, Gondi, Brahui** and others. **Tamil has the oldest continuous literary tradition of any Indian language still in use**
- 3 Two smaller families: **Austro-Asiatic** (Santhali, Mundari, Khasi) and **Tibeto-Burman** (Manipuri, Bodo, Naga languages)

The examinable point about Dravidian, because it is regularly got wrong: the Dravidian languages are a separate family and are NOT derived from Sanskrit. They have borrowed heavily from Sanskrit, which is a different matter. **Tamil in particular has a literary tradition - the Sangam corpus of F5.4 - that is independent of the Sanskrit tradition rather than derived from it.**

The three ancient literary languages, and who used which:

- 1 **SANSKRIT** - the language of the Vedas, the epics, the Puranas, the six schools, and the classical drama of the Gupta age. **The language of ritual and of the learned**
- 2 **PRAKRIT** - the spoken vernaculars. **Ashoka's edicts are in Prakrit; the Jain canon is in Ardhamagadhi Prakrit**
- 3 **PALI** - the language of the **Buddhist Tripitaka**

Notice the pattern, because it is the thread running from F5.3: the reform movements chose the spoken languages and the orthodox tradition kept Sanskrit. That single choice did more for the spread of Buddhism and Jainism than any doctrine.

1.1 The classical languages - the current list

The Government of India confers "classical language" status on languages meeting criteria of antiquity, an independent literary tradition, and a body of ancient literature. The list stood at six for a decade and **was expanded to eleven in October 2024.**

Language	Recognised	Note
Tamil	2004	The first to be recognised. Dravidian, and the oldest continuous literary tradition among living Indian languages - the Sangam corpus of F5.4
Sanskrit	2005	Indo-Aryan. The language of the Vedas and the classical literature
Telugu	2008	Dravidian
Kannada	2008	Dravidian. Telugu and Kannada were recognised in the same year , which is a question in itself
Malayalam	2013	Dravidian

Language	Recognised	Note
Odia	2014	Indo-Aryan, and the first Indo-Aryan language after Sanskrit to be recognised - the point most often tested about it
Marathi	2024	Indo-Aryan
Pali	2024	The Buddhist canonical language of F5.3
Prakrit	2024	The language of Ashoka's edicts and the Jain canon
Assamese	2024	Indo-Aryan
Bengali	2024	Indo-Aryan

The eleven classical languages, with the year of recognition

How to hold this. The **original six in order - Tamil, Sanskrit, Telugu and Kannada, Malayalam, Odia** - then **the five added together in 2024 - Marathi, Pali, Prakrit, Assamese, Bengali**. **Five of the six original ones are of the south or of Sanskrit; four of the five additions are eastern or western Indo-Aryan, plus the two ancient Buddhist and Jain languages.** If a question asks the total, the current answer is **eleven**.

KNOW · THE LITERATURE NOT ALREADY COVERED, IN ONE PASS

Much of the literature in this module has already been placed - **the Vedas and Upanishads in F5.2, the Buddhist and Jain canons in F5.3, and the Sanskrit classics with the Sangam corpus in F5.4**. What remains is the epic and Puranic material and the grammatical tradition.

- The two epics.** The **Ramayana**, attributed to **Valmiki**, and the **Mahabharata**, attributed to **Vyasa**. **The Mahabharata is the longer - conventionally the longest poem in the world - and it contains the BHAGAVAD GITA** as an episode within it, in the Bhishma Parva. Both grew over centuries and are compilations rather than single-author works
- The Puranas.** Eighteen **Mahapuranas**, of which the **Bhagavata, Vishnu, Matsya, Vayu** and **Markandeya** are the ones named. They are the main source for the **theistic devotional religion** that F5.3 identified as the orthodox tradition's answer to the reform movements
- The Dharmashastras.** Legal and social codes: the **Manusmriti** is the best known, with **Yajnavalkya** and **Narada** also named. **These are the texts that state the varna and ashrama order as law**
- Grammar.** **Panini's Ashtadhyayi** is the great Sanskrit grammar and one of the most rigorous analytical works of the ancient world; **Patanjali's Mahabhashya** is its principal commentary. **Tolkappiyam** is the Tamil equivalent, from F5.4
- Fable and lexicon.** The **Panchatantra** of **Vishnu Sharma** and the **Amarakosha** of **Amarasimha**, both from F5.4's Gupta table

The single pairing most often tested from this box: the Bhagavad Gita is part of the MAHABHARATA, not a separate book and not part of the Ramayana.

CHECK YOURSELF · SESSION 1

- Name the two great language families and three languages of each
- Are the Dravidian languages derived from Sanskrit? State the position precisely
- Pair Sanskrit, Prakrit and Pali with what each was used for
- Give the original six classical languages in order of recognition, with years
- Name the five languages added in 2024, and the current total
- Which was the first classical language, and which the first Indo-Aryan after Sanskrit?
- Name the two epics with their attributed authors, and say which is longer
- Where does the Bhagavad Gita sit?
- What are the Puranas a source for, and what do the Dharmashastras state?

2. Session 2 - Ancient Indian Science, Honestly



The real achievements are large. The inflated claims are unnecessary and they cost you credibility. This session separates them.

MUST MASTER · MATHEMATICS - WHERE THE STRONGEST CLAIMS ARE, AND THEY ARE SOLID

This is the field in which ancient and early medieval India made contributions of unambiguous world importance, and the evidence is textual and demonstrable.

- 1 **The DECIMAL PLACE-VALUE SYSTEM with ZERO as a number. The single most consequential Indian contribution to world knowledge.** Place value means a digit's value depends on its position; **zero as a placeholder AND as a number in its own right** is what makes the system work. It reached Europe through Arab intermediaries, which is why the digits are called **Arabic numerals in Europe** though their origin is Indian
- 2 **BRAHMAGUPTA** (7th century) - **the first to give rules for arithmetic with ZERO and with NEGATIVE numbers**, in the **Brahmasphutasiddhanta**. He treated negatives as debts and positives as fortunes, which is a banker's framing you will appreciate. **He also stated, correctly, that division by zero is not defined in the ordinary way** - his own formulation is imperfect but the problem is identified
- 3 **ARYABHATA** (5th century) - the **Aryabhatiya**, from F5.4. Place-value arithmetic, an accurate value of **pi** (given as 3.1416 and described as approximate), trigonometric sine tables, **the earth's rotation on its axis**, and **eclipses explained as shadows** rather than portents
- 4 **The SULBASUTRAS** - appendices to the Vedic Kalpa texts, giving geometric constructions for building sacrificial altars. **They contain what is in effect the Pythagorean relation, stated before Pythagoras**, though as a construction rule rather than as a proved theorem. **State it that way and it is unimpeachable; state it as "Pythagoras's theorem was discovered in India" and you have overclaimed**
- 5 **BHASKARA II** (12th century) - the **Siddhanta Shiromani**, including the **Lilavati** on arithmetic
- 6 **The KERALA SCHOOL** under **Madhava of Sangamagrama** (14th century) - **infinite series for trigonometric functions, anticipating results later associated with Newton and Leibniz**. Genuinely remarkable and under-known

KNOW · ASTRONOMY, MEDICINE AND METALLURGY

ASTRONOMY. Varahamihira's **Brihatsamhita** and **Panchasiddhantika** from F5.4; the **Surya Siddhanta** as the standard astronomical text. **The Jantar Mantar observatories** at Jaipur, Delhi and elsewhere are much later - **early 18th century, built by Sawai Jai Singh II** - and should not be dated to antiquity, which is a common error.

MEDICINE - Ayurveda, and the two names that matter:

- 1 **CHARAKA** - the **Charaka Samhita**, on **general medicine**. Belongs to the **Kushana** period, and F5.4 warned against dating him to the Guptas
- 2 **SUSRUTA** - the **Susruta Samhita**, on **SURGERY**. It describes surgical instruments, procedures, and notably **rhinoplasty - reconstruction of the nose using a flap of skin from the cheek or forehead. This one is genuinely attested in the text and is the responsible version of the "ancient Indian plastic surgery" claim**
- 3 **Sushruta is the surgical text and Charaka the medical one** - that pairing is the question
- 4 **Dhanvantari** is the traditional figure associated with Ayurveda, placed at Chandragupta II's court in the Navaratna tradition

METALLURGY - two solid, demonstrable achievements:

- 1 **The MEHRAULI IRON PILLAR**, Delhi, from F5.4 - **has resisted serious corrosion for about sixteen centuries**, owing to a protective phosphate layer formed by its unusually high phosphorus and low sulphur content. **This is measurable, published metallurgy, not legend**
- 2 **WOOTZ STEEL** - a crucible steel produced in south India and exported west, where it was the basis of the celebrated **Damascus** blades. **A genuine Indian technological export**

And one more solid claim: NAGARJUNA - the name attaches both to the Madhyamaka philosopher of F5.3 and, in tradition, to an alchemist credited with work on metals and mercury. **Treat the alchemical Nagarjuna as traditional attribution rather than established biography.**

BEGINNER MISTAKE · WHAT IS OVERSTATED - AND WHY SAYING SO PROTECTS YOU

Popular writing about ancient Indian science circulates several claims that **are not supported by the evidence**. A candidate who repeats them in an interview loses credibility instantly; one who can distinguish them gains it.

- 1 **Aircraft (vimanas) in antiquity.** The texts cited are either mythological or modern - the "Vaimanika Shastra" is a **twentieth-century** work, not an ancient one. **There is no ancient Indian aeronautics**
- 2 **Stem-cell technology, genetic engineering or test-tube birth** read into epic narratives. **These are literary and mythological passages, not technical descriptions**
- 3 **"Ancient Indians knew the exact speed of light or the precise distance to the sun."** Ancient Indian astronomy contains real quantitative work, **but these specific claims come from strained readings of verses**
- 4 **"Pythagoras's theorem was Indian."** The relation appears in the **Sulbasutras as a construction rule and does predate Pythagoras** - but a construction rule is not a proved theorem, and the responsible statement is the narrower one

The honest summary to carry, and it is a good interview answer. India's demonstrable contributions - the decimal place-value system with zero, Brahmagupta's rules for zero and negatives, Aryabhata's rotating earth and shadow theory of eclipses, Susruta's surgery, the Kerala school's infinite series, wootz steel and the corrosion-resistant iron pillar - are large enough to state confidently without exaggeration. The inflated claims are not merely unnecessary; they crowd out the real ones. That formulation shows judgement, which is what a panel is testing.

2.1 The ancient universities

Centre	Where and when	What to know
Takshashila (Taxila)	Gandhara, from about the 6th century BCE	The earliest. Not a university in the later sense but a cluster of teachers taking pupils individually. Panini and, by tradition, Kautilya and the physician Charaka are associated with it
NALANDA	Bihar; founded by KUMARAGUPTA I in the 5th century CE	The greatest. A residential monastic university with thousands of students and a vast library, described at length by HIUEN TSANG . Sacked around 1200 CE , and F5.3 named that as one of the five causes of Buddhism's decline in India. UNESCO World Heritage Site
Vikramashila	Bihar; Pala dynasty	The second great Pala foundation, a centre of Vajrayana Buddhism. Also destroyed around 1200 CE
Valabhi	Gujarat	A Jain and Buddhist centre, and the site of the Second Jain Council of 512 CE from F5.3
Odantapuri	Bihar; Pala	The third Pala foundation, and traditionally the model for Tibetan monastic organisation
Others	-	Kanchipuram, Ujjain, Somapura (in present-day Bangladesh), Sharada Peeth in Kashmir

The great centres of learning

The two pairings that carry this table: NALANDA was founded by KUMARAGUPTA I and described by HIUEN TSANG; VIKRAMASHILA and ODANTAPURI are PALA foundations. And note the structural point - **the great universities were monastic, which is precisely why their destruction ended a tradition**, the argument F5.3 used to explain why Buddhism vanished from India while Jainism did not.

CHECK YOURSELF · SESSION 2

- 1 What are the two components of the decimal place-value system, and why does zero matter twice?
- 2 What did Brahmagupta establish rules for, and in which work?
- 3 List four results from the Aryabhatiya
- 4 What do the Sulbasutras contain, and what is the responsible way to state it?
- 5 Who was Madhava of Sangamagrama and what did the Kerala school achieve?
- 6 Pair Charaka and Susruta with their subjects, and give the surgical procedure attested in Susruta
- 7 When were the Jantar Mantar observatories built, and by whom?
- 8 Explain why the Mehrauli pillar resists corrosion, and what wootz steel was
- 9 Name four overstated claims and say why distinguishing them matters
- 10 Who founded Nalanda, who described it, and which centres were Pala foundations?

3. Session 3 - Dance ★★★★★

Eight classical forms, eight states. This is the most reliably asked set in the whole cultural half of the module.

START FROM ZERO · THE NATYASHASTRA, AND THE VOCABULARY EVERY DANCE QUESTION USES

The **NATYASHASTRA**, attributed to **BHARATA (Bharata Muni)**, is the foundational treatise on drama, dance and music - conventionally placed between about **200 BCE and 200 CE**. It is sometimes called the **fifth Veda** because of the authority it acquired, and **every classical form in India traces its legitimacy to it**.

The distinctions it draws, and they are asked directly:

- 1 **NRITTA** - **pure dance**: rhythm, movement and pattern, with no story and no meaning
- 2 **NRITYA** - **expressive dance**: movement used to convey meaning and emotion
- 3 **NATYA** - **drama**: dance with narrative, character and speech

RASA - the theory of aesthetic emotion, and the count is a question. There are **NINE rasas (navarasa)**: **shringara** (love), **hasya** (mirth), **karuna** (compassion), **raudra** (anger), **veera** (heroism), **bhayanaka** (fear), **bibhatsa** (disgust), **adbhuta** (wonder) and **shanta** (peace). **Shanta is traditionally the ninth and latest addition** - the Natyashastra's original scheme has eight, which is why some sources say eight and the standard modern answer is nine.

ABHINAYA - the four modes of expression: **angika** (body and limb), **vachika** (speech and song), **aharya** (costume and ornament) and **sattvika** (inner or psychological). And **TANDAVA** is the vigorous, masculine aspect of dance, **LASYA** the gentle, graceful one.

One more text worth the name: the ABHINAYA DARPANA of NANDIKESHVARA, a later manual on gesture, and the source of much of what is taught as hand-gesture technique today.

3.1 The eight classical dances

The Sangeet Natak Akademi recognises **EIGHT classical dance forms**. Some sources say nine by including **Chhau**, which the Akademi classifies as a folk or martial tradition - **if a question asks the number of classical dances, answer eight, and know why nine appears**.

Dance	State	What to know
BHARATANATYAM	Tamil Nadu	The oldest of the classical forms , developed from the Sadir of the temple devadasi tradition. Revived in the twentieth century by Rukmini Devi Arundale and E. Krishna Iyer , which is the standard question about it. Characteristic half-sitting araimandi posture; usually solo (ekaharya) ; sometimes called the fire dance for its linear, flame-like lines
KATHAK	Uttar Pradesh	The ONLY classical dance form of NORTHERN India - the single most useful fact in the table. Name from the Kathakas , hereditary storytellers. Developed in both temple and Mughal court settings, so it is the one form showing genuine Hindu-Muslim synthesis . Known for fast footwork and spins (chakkars) . Gharanas: LUCKNOW, JAIPUR and BANARAS (with Raigarh sometimes added). Birju Maharaj is the modern name
KATHAKALI	Kerala	A dance-DRAMA , not a solo dance - the name means "story-play". Famous for elaborate facial make-up (chutti) and towering costume, and for stories from the epics . Traditionally performed by men
KUCHIPUDI	Andhra Pradesh	Traced to SIDDHENDRA YOGI and the Bhagavata Mela tradition of dance-drama. Its signature item is the TARANGAM - dancing on the rim of a brass plate , often balancing a pot of water. That single detail is the reliable identifier
ODISSI	Odisha	Associated with the Jagannath temple and the Mahari (temple dancer) and Gotipua (boy dancer) traditions. Its signature is the TRIBHANGA - the thrice-bent posture of head, torso and knees - with the square chauka stance. Kelucharan Mohapatra is the modern name

Dance	State	What to know
MANIPURI	Manipur	Centred on Ras Leela and the Radha-Krishna theme. Gentle, flowing and rounded, with restrained facial expression - the opposite emphasis to Kathakali. Associated forms: Sankirtana and the drum dance Pung Cholom
MOHINIYATTAM	Kerala	Solo, and traditionally female - the name comes from mohini , the enchantress. Emphasises lasya , the graceful mode, with swaying torso movement. Codified under Swathi Thirunal of Travancore. Kerala therefore has TWO classical forms - Kathakali and Mohiniyattam - which is a standard trap
SATTRIYA	Assam	Introduced by the Vaishnava saint SRIMANTA SANKARADEVA in the sixteenth century, and performed in the monasteries called sattras , from which it takes its name. The most recently recognised as classical, in 2000 - which is the fact usually asked

The eight classical dances - the state is the question

HIGH YIELD · THE FOUR DANCE FACTS MOST LIKELY TO DECIDE A QUESTION

- 1 **Kathak is the only northern classical form.** Every other one of the eight is southern, eastern or north-eastern
- 2 **Kerala has TWO - Kathakali and Mohiniyattam.** A question offering both as options for "the classical dance of Kerala" is testing whether you know there are two, and the discriminator is that **Kathakali is a male dance-drama and Mohiniyattam a solo female form**
- 3 **Sattriya was introduced by Sankaradeva and recognised as classical only in 2000**
- 4 **Tarangam belongs to Kuchipudi and tribhanga to Odissi** - two signature details, two different states

And the counting point: EIGHT classical forms per the Sangeet Natak Akademi. Chhau is the ninth candidate and is classified as folk or martial, though it carries UNESCO Intangible Cultural Heritage listing in its own right.

KNOW · FOLK DANCES AND THE UNESCO INTANGIBLE LIST

Folk dances are pure state-pairing questions. Learn the well-known ones and stop.

- 1 **Punjab - Bhangra** (men), **Giddha** (women) · **Gujarat - Garba, Dandiya Raas** · **Rajasthan - Ghoomar, Kalbelia** · **Maharashtra - Lavani, Tamasha**
- 2 **Assam - Bihu** · **Odisha, Jharkhand and West Bengal - Chhau** · **Karnataka - Yakshagana, Dollu Kunitha** · **Kerala - Theyyam, Ottamthullal**
- 3 **Mizoram - Cheraw** (the bamboo dance) · **Tripura - Hojagiri** · **Jammu and Kashmir - Rouf** · **Uttar Pradesh - Nautanki** · **Bihar - Bidesia** · **Andhra Pradesh - Burrakatha**

UNESCO's Intangible Cultural Heritage list is the other frame these questions take, and India's entries include: Kutiyattam (Sanskrit theatre, Kerala), **Vedic chanting, Ramlila, Ramman** (Uttarakhand), **Chhau, Kalbelia, Mudi yettu** (Kerala), **Buddhist chanting of Ladakh, Sankirtana** (Manipur), the **Thatheras' brass and copper craft** (Punjab), **Yoga, Kumbh Mela, Durga Puja in Kolkata**, and the **Garba of Gujarat** (2023).

Two of those are worth separating because they are recent and therefore fashionable in question-setting: Durga Puja of Kolkata and the Garba of Gujarat are the two most recent Indian additions.

4. Session 4 - Music and Painting ★★★★★

Two music systems from one root, and a painting tradition that splits by court. Both are pairing exercises.

MUST MASTER · HINDUSTANI AND CARNATIC - ONE ROOT, TWO SYSTEMS

Indian classical music descends from the SAMA VEDA (F5.2) through the Natyashastra and the great medieval treatise the SANGITA RATNAKARA of SARANGADEVA (13th century), which is pre-bifurcation - it is common ancestor to both systems, and that is why it is asked. The division into two systems dates from roughly the **thirteenth to fourteenth centuries**, as Persian and Central Asian influence reshaped music in the north.

Both systems share the two fundamental concepts: RAGA - the melodic framework - and TALA - the rhythmic cycle. What differs is emphasis and organisation.

Feature	HINDUSTANI (north)	CARNATIC (south)
Emphasis	Improvisation. An extended, unmetred alap explores the raga before rhythm enters	Composition. The kriti is central, and improvisation happens within and around a composed piece
Organising scheme	10 THAATS , as systematised by V. N. Bhatkhande	72 MELAKARTAS , as systematised by Venkatamakhin
Principal forms	Khayal (the dominant form), Dhrupad (the older, austere form), Thumri , Tarana , Dhamar , Tappa	Varnam , Kriti , and Ragam-Tanam-Pallavi as the elaborate improvisational centrepiece
Instruments	Sitar, sarod, sarangi, santoor, shehnai, tabla	Veena, mridangam, violin, ghatam, nadaswaram, kanjira
Transmission	Gharanas - hereditary schools such as Gwalior, Kirana, Jaipur-Atrauli, Agra	Composer-lineages and the guru-shishya tradition
Names to know	Tansen at Akbar's court; later Bhimsen Joshi, Ravi Shankar	Purandara Dasa , called the father of Carnatic music ; then the TRINITY

Hindustani against Carnatic - the comparison asked

THE CARNATIC TRINITY, and this is the highest-value music fact: TYAGARAJA, MUTHUSWAMI DIKSHITAR and SYAMA SASTRI, all of the eighteenth to nineteenth centuries. **Do not confuse the Trinity with Purandara Dasa, who is the "father of Carnatic music" but is NOT one of the three** - that is exactly how the question is built.

JARGON DECODER · THE FOUR CLASSES OF INSTRUMENT - A NATYASHASTRA SCHEME STILL USED

The Natyashastra classifies instruments by **how the sound is produced**, and the four Sanskrit names appear verbatim in questions. **As an engineer you will notice this is the same basis as the modern Hornbostel-Sachs classification, arrived at some two thousand years earlier.**

- 1 TATA VADYA - stringed** (chordophones): **veena, sitar, sarod, sarangi, violin**
- 2 SUSHIRA VADYA - wind** (aerophones): **flute or bansuri, shehnai, nadaswaram**
- 3 AVANADDHA VADYA - percussion with a stretched membrane** (membranophones): **tabla, mridangam, pakhawaj, dholak**
- 4 GHANA VADYA - solid percussion without a membrane** (idiophones): **manjira, jhanj, ghatam, jaltarang**

The distinction that decides the question is between **AVANADDHA** and **GHANA** - both are percussion, but **avanaddha** has a stretched skin and **ghana** does not. So **tabla** and **mridangam** are **avanaddha**; **manjira**, **cymbals** and the **clay-pot ghatam** are **ghana**.

4.1 The painting schools

Held back from F5.5 deliberately, because these schools belong to the courts and communities that produced them rather than to architecture.

School	Where and when	What to know
MUGHAL	Imperial court, 16th-18th centuries	AKBAR founded the atelier, fusing Persian technique with Indian subject matter - the illustrated Hamzanama and Razmnama ; artists Basawan and Daswanth . JAHANGIR is the peak - a connoisseur, and his reign favours naturalism, portraiture and studies of birds and animals , with USTAD MANSUR the great natural-history painter and Abul Hasan the portraitist. SHAH JAHAN brings formality and jewel-like finish. AURANGZEB withdrew patronage, and the dispersal of his artists is what enriched the Rajput and Deccani courts - the decline of one school being the making of others
RAJASTHANI / RAJPUT	Rajasthan courts	Sub-schools by court: Mewar, Bundi, Kota (hunting scenes), Kishangarh (famous for the Bani Thani female type), Marwar, Amber-Jaipur . Themes: Krishna and the Bhagavata, Ragamala (paintings of musical modes) and Barahmasa (the twelve months)
PAHARI	Himalayan foothills, Himachal and Jammu	BASOHLI - the early phase, bold colour, and beetle-wing green used for jewellery . KANGRA - the mature phase, lyrical and refined , associated with the family of Nainsukh and with the Gita Govinda and Krishna themes. Also Guler, Chamba, Garhwal
DECCANI	Ahmadnagar, Bijapur, Golconda, later Hyderabad	Independent of Mughal work and often earlier in its use of rich colour and fantasy; Golconda and Bijapur are the names to hold
Company style	18th-19th centuries	Indian artists working for British patrons , producing hybrid Indo-European work - natural-history plates, costume studies, topography
TANJORE	Thanjavur, Tamil Nadu	Gold foil over a gesso ground , with gem inlay; devotional subjects, chiefly Krishna. The gold leaf is the identifier

The miniature and later painting schools

KNOW · THE FOLK AND TRIBAL PAINTING TRADITIONS - PURE STATE PAIRINGS

These appear in general-awareness questions as often as in history ones, and they are the cheapest facts in the chapter.

- 1 MADHUBANI (Mithila) - BIHAR.** Line-based, filled with natural dyes; no space left blank
- 2 WARLI - MAHARASHTRA.** Tribal; white rice-paste figures built from triangles and circles on an earth-coloured ground
- 3 PATTACHITRA - ODISHA** (and West Bengal). Cloth-based scroll painting, Jagannath themes
- 4 KALAMKARI - ANDHRA PRADESH.** Pen-and-dye work on cotton; the two centres are **Srikalahasti** (freehand, temple narratives) and **Machilipatnam** (block-printed)
- 5 PHAD - RAJASTHAN.** Long narrative scrolls of folk deities
- 6 GOND - MADHYA PRADESH.** Tribal, built from dots and dashes
- 7 KALIGHAT - WEST BENGAL.** Nineteenth-century Kolkata, bold brush outlines and social satire
- 8 CHERIYAL scrolls - TELANGANA · SAURA - ODISHA · SOHRAI-KHOVAR - JHARKHAND · AIPAN - UTTARAKHAND · PITHORA - GUJARAT and MADHYA PRADESH · THANGKA - Buddhist, Ladakh and Sikkim**

The four that are asked far more than the rest: MADHUBANI-Bihar, WARLI-Maharashtra, PATTACHITRA-Odisha, KALAMKARI-Andhra Pradesh. Learn those four cold and treat the remainder as recognition only.

ACTION · WHERE THIS LEAVES YOU, AND WHAT F5.7 DOES

The cultural half of Module 5 is now complete. Between F5.3, F5.5 and this chapter you hold religion and philosophy, architecture and sculpture, and now languages, literature, science, dance, music and painting. **F5.1's five-slot template is full for the ancient period on every slot.**

F5.7 turns to MEDIEVAL INDIA - the Delhi Sultanate and its five dynasties, Vijayanagara and the Bahmanis, the Mughals as a political history, the Marathas, and the **Bhakti and Sufi movements**, which are the medieval continuation of the religious argument F5.3 began. **Indo-Islamic architecture is taken there rather than in F5.5**, with the politics that produced it - so the Qutb complex, Fatehpur Sikri, the Taj Mahal and the provincial styles all belong to that chapter.

Then F5.8 and F5.9 take modern India - company rule to 1857, the rebellions, and the freedom movement with the labour movement that runs alongside it - and **F5.10 closes the module with a master chronology and a consolidated facts sheet**, on the model of F1.13 and F3.10.

5. Graded Practice - 42 Questions ★★★★★

5.1 Level 1 - Recognition

Q. 1. The Natyashastra is attributed to

(a) Bharata (b) Nandikeshvara (c) Sarangadeva (d) Abhinavagupta

Q. 2. Bharatanatyam is the classical dance of

(a) Kerala (b) Tamil Nadu (c) Andhra Pradesh (d) Odisha

Q. 3. The only classical dance form of northern India is

(a) Kathakali (b) Manipuri (c) Kathak (d) Odissi

Q. 4. Kathakali belongs to

(a) Karnataka (b) Manipur (c) Tamil Nadu (d) Kerala

Q. 5. Kuchipudi is the classical dance of

(a) Andhra Pradesh (b) Odisha (c) Assam (d) Kerala

Q. 6. Odissi belongs to

(a) Assam (b) Odisha (c) West Bengal (d) Manipur

Q. 7. Sattriya is the classical dance of

(a) Tripura (b) Manipur (c) Assam (d) Nagaland

Q. 8. Mohiniyattam belongs to

(a) Tamil Nadu (b) Andhra Pradesh (c) Karnataka (d) Kerala

Q. 9. The first language to be granted classical language status was

(a) Tamil (b) Sanskrit (c) Telugu (d) Odia

Q. 10. The Bhagavad Gita forms part of the

(a) Ramayana (b) Mahabharata (c) Rig Veda (d) Bhagavata Purana

Q. 11. The Aryabhatiya was written by

(a) Brahmagupta (b) Varahamihira (c) Aryabhata (d) Bhaskara II

Q. 12. The Susruta Samhita deals principally with

(a) general medicine (b) grammar (c) astronomy (d) surgery

Q. 13. Nalanda Mahavihara was founded by

(a) Kumaragupta I (b) Ashoka (c) Kanishka (d) Harshavardhana

Q. 14. Which one of the following is one of the Trinity of Carnatic music?

(a) Tansen (b) Tyagaraja (c) Amir Khusrau (d) Purandara Dasa

Q. 15. Madhubani painting belongs to

(a) Odisha (b) Maharashtra (c) Bihar (d) Bengal

Q. 16. The number of classical dance forms recognised by the Sangeet Natak Akademi is

(a) six (b) eleven (c) nine (d) eight

5.2 Level 2 - Understanding

Q. 17. The number of rasas in the standard modern scheme is

(a) nine (b) four (c) ten (d) sixteen

Q. 18. Sattriya was introduced by

(a) Siddhendra Yogi (b) Srimanta Sankaradeva (c) Swathi Thirunal (d) Rukmini Devi Arundale

Q. 19. The principal gharanas of Kathak are

(a) Gwalior, Kirana and Agra (b) Mewar, Bundi and Kota (c) Lucknow, Jaipur and Banaras (d) Basohli, Guler and Kangra

Q. 20. The tarangam, danced on the rim of a brass plate, belongs to

(a) Odissi (b) Manipuri (c) Bharatanatyam (d) Kuchipudi

Q. 21. The tribhanga, or thrice-bent posture, is characteristic of

(a) Odissi (b) Kathak (c) Mohiniyattam (d) Sattriya

Q. 22. The relationship between the Dravidian languages and Sanskrit is that

(a) the Dravidian languages descend from Sanskrit (b) they form a separate family and are not derived from Sanskrit, though they have borrowed from it (c) Sanskrit descends from Tamil (d) they belong to the Austro-Asiatic family

Q. 23. Brahmagupta is credited with

(a) the first Sanskrit grammar (b) the discovery of the zero symbol alone (c) the first rules for arithmetic with zero and with negative numbers (d) the theory of rasa

Q. 24. The Sulbasutras contain

(a) rules of Sanskrit prosody (b) astronomical tables only (c) medical prescriptions (d) geometric constructions for building sacrificial altars, including what amounts to the Pythagorean relation as a construction rule

Q. 25. The Mehrauli iron pillar has resisted corrosion for about sixteen centuries because of

(a) a protective phosphate layer arising from unusually high phosphorus and low sulphur content (b) a coating of gold applied in the Gupta period (c) the dry climate of Delhi alone (d) a modern chemical treatment

Q. 26. The four classes of musical instrument in the Natyashastra scheme are

(a) raga, tala, laya and sruti (b) tata, sushira, avanaddha and ghana (c) khayal, dhrupad, thumri and tarana (d) nritta, nritya, natya and abhinaya

Q. 27. The number of melakartas in the Carnatic system is

(a) ten (b) eighteen (c) seventy-two (d) one hundred and eight

Q. 28. Mughal painting reached its peak, with an emphasis on naturalism and studies of birds and animals, under

(a) Akbar (b) Aurangzeb (c) Shah Jahan (d) Jahangir

Q. 29. Warli painting belongs to

(a) Maharashtra (b) Bihar (c) Odisha (d) Telangana

Q. 30. The Sangita Ratnakara of Sarangadeva is important because it

(a) is the earliest work on Carnatic music alone (b) predates the division into Hindustani and Carnatic and is common ancestor to both (c) was written at Akbar's court (d) is a treatise on painting

5.3 Level 3 - Recruitment Test standard**Q. 31. Match List-I with List-II and select the correct answer using the code below:**

List-I (Dance)	List-II (State)
A. Kuchipudi	1. Assam
B. Sattriya	2. Odisha
C. Mohiniyattam	3. Andhra Pradesh
D. Odissi	4. Kerala

(a) A-3 B-4 C-1 D-2 (b) A-1 B-3 C-4 D-2 (c) A-3 B-1 C-4 D-2 (d) A-2 B-1 C-4 D-3

Q. 32. Which one of the following statements about the classical dances is NOT correct?

(a) Kathak is the only classical dance form of northern India (b) Sattriya was introduced by Srimanta Sankaradeva (c) Kathakali is a dance-drama of Kerala (d) Bharatanatyam originated in Andhra Pradesh

Q. 33. Match List-I with List-II and select the correct answer using the code below:

List-I (Painting school)	List-II (Region)
A. Kangra	1. Deccan
B. Kishangarh	2. Himachal hills
C. Golconda	3. Tamil Nadu
D. Tanjore	4. Rajasthan

(a) A-2 B-4 C-1 D-3 (b) A-4 B-2 C-1 D-3 (c) A-2 B-1 C-4 D-3 (d) A-1 B-4 C-2 D-3

Q. 34. Consider the following statements about the classical languages: 1. Tamil was the first to be granted the status. · 2. Odia was the first Indo-Aryan language after Sanskrit to be recognised. · 3. The list currently stands at six. Which are correct?

(a) 1 only (b) 1 and 2 only (c) 2 and 3 only (d) 1, 2 and 3

Q. 35. Match List-I with List-II and select the correct answer using the code below:

List-I (Author)	List-II (Work)
A. Aryabhata	1. Brahmasphutasiddhanta
B. Brahmagupta	2. Siddhanta Shiromani
C. Varahamihira	3. Aryabhatiya
D. Bhaskara II	4. Brihatsamhita

(a) A-3 B-4 C-1 D-2 (b) A-1 B-3 C-4 D-2 (c) A-3 B-1 C-4 D-2 (d) A-2 B-1 C-3 D-4

Q. 36. Which one of the following statements about ancient Indian science is NOT correct?

(a) The decimal place-value system with zero is an Indian contribution transmitted to Europe through Arab intermediaries (b) Brahmagupta gave rules for arithmetic with zero and with negative numbers (c) The Susruta Samhita describes reconstruction of the nose using a skin flap (d) The Vaimanika Shastra is an ancient text describing aircraft

Q. 37. Which one of the following best describes Carnatic music?

(a) it is composition-centred, organised on seventy-two melakartas, with Tyagaraja, Muthuswami Dikshitar and Syama Sastri as its Trinity (b) it is organised on ten thaats and centres on the khayal (c) it is transmitted through gharanas such as Gwalior and Kirana (d) it was systematised by V. N. Bhatkhande

Q. 38. Arrange the following in correct chronological order, earliest first: 1. Sangita Ratnakara of Sarangadeva · 2. Sama Veda · 3. the Trinity of Carnatic music · 4. Natyashastra

(a) 2-1-4-3 (b) 2-4-1-3 (c) 4-2-1-3 (d) 2-4-3-1

Q. 39. Which one of the following statements about Nalanda is NOT correct?

(a) It was founded by Kumaragupta I (b) It was described at length by Hiuen Tsang (c) It was founded by Ashoka in the third century BCE (d) It was a residential monastic university

Q. 40. Match List-I with List-II and select the correct answer using the code below:

List-I (Folk painting)	List-II (State)
A. Madhubani	1. Odisha
B. Warli	2. Bihar
C. Pattachitra	3. Andhra Pradesh
D. Kalamkari	4. Maharashtra

(a) A-1 B-4 C-2 D-3 (b) A-2 B-1 C-4 D-3 (c) A-4 B-2 C-1 D-3 (d) A-2 B-4 C-1 D-3

Q. 41. Consider the following statements about Kathak: 1. It is the only classical dance form of northern India. · 2. Its principal gharanas include Lucknow, Jaipur and Banaras. · 3. It developed in both temple and Mughal court settings and shows Hindu-Muslim synthesis. Which are correct?

(a) 1, 2 and 3 (b) 1 and 2 only (c) 2 and 3 only (d) 1 only

Q. 42. Match List-I with List-II and select the correct answer using the code below:

List-I (Instrument)	List-II (Class)
A. Veena	1. Sushira
B. Flute	2. Ghana
C. Mridangam	3. Tata
D. Manjira	4. Avanaddha

(a) A-1 B-3 C-4 D-2 (b) A-3 B-1 C-4 D-2 (c) A-3 B-4 C-1 D-2 (d) A-4 B-1 C-3 D-2

6. Answers and Explanations

6.1 Level 1 - answers

Q	Ans	Why, and why the others fail
1	a	Bharata (Bharata Muni) , conventionally 200 BCE to 200 CE. (b) Nandikeshvara wrote the later Abhinaya Darpana on gesture; (c) Sarangadeva the Sangita Ratnakara on music; (d) Abhinavagupta a commentary. All four are real names in this field, which is what makes the question work
2	b	Tamil Nadu - the oldest of the eight, from the Sadir of the devadasi tradition, revived by Rukmini Devi Arundale and E. Krishna Iyer
3	c	Kathak - the single most useful fact in the dance table. Every other classical form is southern, eastern or north-eastern
4	d	Kerala - a male dance-DRAMA with the elaborate chutti make-up. Note that Kerala has TWO classical forms , Kathakali and Mohiniyattam
5	a	Andhra Pradesh , traced to Siddhendra Yogi , with the tarangam on a brass plate as its signature
6	b	Odisha - the tribhanga posture, the Jagannath temple connection, and Kelucharan Mohapatra as the modern name
7	c	Assam - introduced by Srimanta Sankaradeva , performed in sattras , and recognised as classical only in 2000
8	d	Kerala - solo, traditionally female, emphasising lasya , codified under Swathi Thirunal . (a) Tamil Nadu is Bharatanatyam - the trap here is Kerala's two forms
9	a	Tamil, in 2004 - the first of all. (b) Sanskrit followed in 2005 , which is the intended near-miss, and (d) Odia in 2014 was the first Indo-Aryan language after Sanskrit
10	b	The Mahabharata , in the Bhishma Parva. (a) The Ramayana is the other epic, attributed to Valmiki , and the Gita has nothing to do with it
11	c	Aryabhata , 5th century. (a) Brahmagupta wrote the Brahmasphutasiddhanta , (b) Varahamihira the Brihatsamhita , (d) Bhaskara II the Siddhanta Shiromani - four authors, four works, and Q35 tests the same set as a matching item
12	d	Surgery - and it describes rhinoplasty using a skin flap , which is the responsible version of the "ancient Indian plastic surgery" claim. (a) General medicine is CHARAKA , and that pairing is the question
13	a	Kumaragupta I , in the 5th century, from F5.4. Its sacking around 1200 CE is one of the five causes of Buddhism's decline in F5.3
14	b	Tyagaraja , with Muthuswami Dikshitar and Syama Sastri . (d) Purandara Dasa is the "father of Carnatic music" but is NOT one of the Trinity - the most carefully built distractor in the music topic. (a) Tansen belongs to Akbar's court and the Hindustani tradition

Q	Ans	Why, and why the others fail
15	c	Bihar - the Mithila region. Line-based work in natural dyes with no blank space left
16	d	Eight. (c) Nine appears because some sources add Chhau , which the Akademi classifies as folk or martial - so nine is the informed wrong answer, and knowing why it is offered is the point

6.2 Level 2 - answers

Q	Ans	Why, and why the others fail
17	a	Nine (navarasa) in the standard modern scheme. The Nattyashastra's original list has eight, with shanta added later - which is why some sources say eight, and why the question specifies the modern scheme
18	b	Srimanta Sankaradeva , the Assamese Vaishnava saint. (a) Siddhendra Yogi is Kuchipudi, (c) Swathi Thirunal codified Mohiniyattam, (d) Rukmini Devi Arundale revived Bharatanatyam - four founders or revivers, four different forms
19	c	Lucknow, Jaipur and Banaras. (a) Gwalior, Kirana and Agra are HINDUSTANI MUSIC gharanas - the same word used in a different art, which is the trap. (d) Basohli, Guler and Kangra are PAHARI PAINTING schools
20	d	Kuchipudi - dancing on the rim of a brass plate, often balancing a water pot. A single unmistakable identifier
21	a	Odissi - the thrice-bent posture of head, torso and knees, with the square chauka stance
22	b	A separate family, not derived from Sanskrit, though heavily borrowing from it. (a) is the standard error. Tamil's Sangam literature is an independent tradition rather than a derived one, which is why Tamil was the first language recognised as classical
23	c	The first rules for arithmetic with zero and with negative numbers , in the Brahmasphutasiddhanta - he framed negatives as debts and positives as fortunes. (b) understates it: the achievement is the RULES, not merely a symbol
24	d	Geometric constructions for altars , containing the Pythagorean relation as a construction rule. The responsible statement is the narrow one - a construction rule is not a proved theorem, and "Pythagoras's theorem was Indian" overclaims
25	a	A protective phosphate layer from high phosphorus and low sulphur - measurable, published metallurgy rather than legend. (c) climate alone is insufficient and is the popular half-answer
26	b	Tata (string), sushira (wind), avanaddha (stretched membrane) and ghana (solid). (a) lists musical concepts and (d) the Nattyashastra's dance categories - three sets of four from the same chapter, offered together
27	c	Seventy-two melakartas , systematised by Venkatamakhin. (a) Ten is the number of HINDUSTANI THAATS , systematised by Bhatkhande - the two schemes are the standard pair to confuse
28	d	Jahangir - a connoisseur, whose reign favoured naturalism and natural history, with Ustad Mansur its great painter of birds and animals. (a) Akbar founded the atelier; (b) Aurangzeb withdrew patronage , and the dispersal of his artists enriched the Rajput and Deccani courts
29	a	Maharashtra - tribal work in white rice paste, figures built from triangles and circles
30	b	It predates the bifurcation and is common ancestor to both systems , which is exactly why it is asked. Sarangadeva is 13th century, and the split into Hindustani and Carnatic follows

6.3 Level 3 - answers

Q	Ans	Why, and why the others fail
31	c	A-3 Kuchipudi-Andhra Pradesh · B-1 Sattriya-Assam · C-4 Mohiniyattam-Kerala · D-2 Odissi-Odisha. Knowing only Sattriya-Assam and Odissi-Odisha eliminates every wrong code
32	d	(d) is false. Bharatanatyam is TAMIL NADU; Andhra Pradesh is Kuchipudi . (a), (b) and (c) are all correct, and (a) - Kathak as the only northern form - is the fact most worth carrying out of the session
33	a	A-2 Kangra-Himachal hills (Pahari) · B-4 Kishangarh-Rajasthan · C-1 Golconda-Deccan · D-3 Tanjore-Tamil Nadu. Kishangarh is where the Bani Thani female type comes from, and Tanjore is identified by its gold foil
34	b	1 and 2 only. Statement 3 is false and is the live-data point of the chapter: the list was expanded from six to ELEVEN in October 2024 , adding Marathi, Pali, Prakrit, Assamese and Bengali . Most printed guides still say six
35	c	A-3 Aryabhata-Aryabhatiya · B-1 Brahmagupta-Brahmasphutasiddhanta · C-4 Varahamihira-Brihatsamhita · D-2 Bhaskara II-Siddhanta Shiromani. The same four names as Q11, tested as a matching item - which is how the paper checks whether a set is held or merely recognised
36	d	(d) is false, and it is the chapter's honesty point. The Vaimanika Shastra is a TWENTIETH-CENTURY work, not an ancient text , and there is no ancient Indian aeronautics. (a), (b) and (c) are all solid, demonstrable claims - which is the argument: the real achievements are large enough not to need inflating
37	a	Composition-centred, seventy-two melakartas, and the Trinity. (b), (c) and (d) all describe HINDUSTANI music - ten thaats, the khayal, gharanas such as Gwalior and Kirana, and Bhatkhande's systematisation. Three of the four options belong to the other system, which is the cleanest possible test of the comparison table
38	b	2-4-1-3: the Sama Veda (Vedic period), the Natyashastra (c. 200 BCE-200 CE), the Sangita Ratnakara (13th century), and the Carnatic Trinity (18th-19th centuries). The sequence is the whole descent of Indian classical music in one question
39	c	(c) is false. Nalanda was founded by Kumaragupta I in the fifth century CE, not by Ashoka in the third century BCE. (a), (b) and (d) are all true. Ashoka is offered because he is the default answer for anything ancient and Buddhist - the same reflex that makes candidates call the Barabar caves Buddhist in F5.5
40	d	A-2 Madhubani-Bihar · B-4 Warli-Maharashtra · C-1 Pattachitra-Odisha · D-3 Kalamkari-Andhra Pradesh. These four are asked far more than the rest of the folk traditions and should be known cold
41	a	All three are correct. Kathak is the only northern classical form, its gharanas are Lucknow, Jaipur and Banaras , and it is the one form showing genuine Hindu-Muslim synthesis , having developed in temple and Mughal court settings alike
42	b	A-3 Veena-tata (string) · B-1 Flute-sushira (wind) · C-4 Mridangam-avanaddha (stretched membrane) · D-2 Manjira-ghana (solid). The discriminator is between avanaddha and ghana: both are percussion, but avanaddha has a stretched skin and ghana does not

7. One-Minute Revision

ONE-MINUTE REVISION · CHAPTER F5.6 - TEXTS, SCIENCE, DANCE, MUSIC, PAINTING

WHAT THIS CHAPTER IS almost entirely **two-column PAIRINGS** - language to status, scientist to work, dance to state, school to patron · the one place needing judgement rather than memory is **ancient science, where popular writing overstates the evidence**

LANGUAGE FAMILIES INDO-ARYAN north, from Vedic - Sanskrit, Prakrit, Pali → Hindi, Bengali, Marathi, Gujarati, Punjabi, Odia, Assamese; ~**three-quarters of Indians** · **DRAVIDIAN** south, a **SEPARATE family NOT derived from Sanskrit** though borrowing heavily - Tamil, Telugu, Kannada, Malayalam · **Tamil has the oldest continuous literary tradition among living Indian languages** · also **Austro-Asiatic** (Santhali, Khasi) and **Tibeto-Burman** (Manipuri, Bodo)

THE THREE ANCIENT LITERARY LANGUAGES SANSKRIT Vedas, epics, Puranas, classical drama - ritual and the learned · **PRAKRIT** the vernaculars - **Ashoka's edicts** and the **Jain canon (Ardhamagadhi)** · **PALI** the **Buddhist Tripitaka** · **the thread from F5.3: the reform movements chose the spoken languages and the orthodoxy kept Sanskrit, and that did more for their spread than any doctrine**

CLASSICAL LANGUAGES - LIVE DATA, changed October 2024 from **SIX** to **ELEVEN** original six in order: **TAMIL 2004 (first)**, **SANSKRIT 2005**, **TELUGU** and **KANNADA both 2008**, **MALAYALAM 2013**, **ODIA 2014 (first Indo-Aryan after Sanskrit)** · added 2024: **MARATHI, PALI, PRAKRIT, ASSAMESE, BENGALI** · most printed guides still say six - if both are offered, eleven is current

LITERATURE NOT ALREADY PLACED RAMAYANA-VALMIKI, MAHABHARATA-VYASA · the **Mahabharata is the longer and CONTAINS THE BHAGAVAD GITA** (Bhishma Parva) - **not the Ramayana** · **18 Mahapuranas** - Bhagavata, Vishnu, Matsya - the source for **devotional theism** · **Dharmashastras** - **MANUSMRITI**, Yajnavalkya, Narada - state varna and ashrama as law · grammar **Panini's Ashtadhyayi** with **Patanjali's Mahabhashya**; Tamil equivalent **Tolkappiyam**

MATHEMATICS - the strongest and most solid claims DECIMAL PLACE VALUE WITH ZERO - the single most consequential Indian contribution, reaching Europe via **Arab intermediaries**, hence "Arabic numerals" · **BRAHMAGUPTA (7th c)**, **Brahmasphutasiddhanta - FIRST RULES FOR ZERO AND NEGATIVE NUMBERS**, negatives as debts · **ARYABHATA (5th c)**, **Aryabhatiya - place value, pi as 3.1416 stated as approximate, sine tables, THE EARTH ROTATES, ECLIPSES AS SHADOWS** · **SULBASUTRAS** - altar geometry containing the **Pythagorean relation as a CONSTRUCTION RULE, which does predate Pythagoras** · **BHASKARA II (12th c)**, **Siddhanta Shiromani incl. Lilavati** · **KERALA SCHOOL under MADHAVA OF SANGAMAGRAMA (14th c) - INFINITE SERIES** anticipating Newton and Leibniz

ASTRONOMY, MEDICINE, METALLURGY Varahamihira - Brihatsamhita, Panchasiddhantika; Surya Siddhanta · **JANTAR MANTAR is EARLY 18th CENTURY by SAWAI JAI SINGH II - NOT ancient** · **CHARAKA = general medicine, KUSHANA period; SUSRUTA = SURGERY, incl. RHINOPLASTY with a skin flap** · **Dhanvantari** traditional Ayurveda figure · **MEHRAULI IRON PILLAR - a PHOSPHATE layer from high phosphorus and low sulphur, published metallurgy not legend** · **WOOTZ STEEL** exported west, the basis of **Damascus blades**

WHAT IS OVERSTATED - and saying so protects you no ancient Indian aircraft; the "Vaimanika Shastra" is TWENTIETH CENTURY · no stem cells or genetic engineering in the epics · no exact speed of light · **"Pythagoras's theorem was Indian" overclaims - it is a construction rule, not a proved theorem - the honest summary: the demonstrable contributions are large enough to state confidently, and the inflated ones crowd out the real ones**

THE UNIVERSITIES TAKSHASHILA earliest, Gandhara, from ~6th c BCE, individual teachers - **Panini**, by tradition **Kautilya** and **Charaka** · **NALANDA founded by KUMARAGUPTA I, described by HIUEN TSANG, sacked c. 1200 CE, UNESCO** · **VIKRAMASHILA** and **ODANTAPURI** are **PALA foundations** · **VALABHI** Jain and Buddhist, site of the **Second Jain Council 512 CE** · the universities were **MONASTIC**, which is why destroying them ended a tradition - the **F5.3 argument**

NATYASHASTRA by **BHARATA**, c. 200 BCE-200 CE, the "fifth Veda" · **NRITTA** pure dance · **NRITYA** expressive · **NATYA** drama · **NINE RASAS (navarasa)**: shringara, hasya, karuna, raudra, veera, bhayanaka, bibhatsa, adbhuta, **shanta (the later ninth, so older sources say eight)** · **ABHINAYA** **four modes**: angika, vachika, aharya, sattvika · **TANDAVA** vigorous, **LASYA** graceful · **Abhinaya Darpana** by **NANDIKESHVARA**

THE EIGHT CLASSICAL DANCES - EIGHT per the **Sangeet Natak Akademi BHARATANATYAM** **Tamil Nadu** - oldest, from **Sadir/devadasi**, revived by **RUKMINI DEVI ARUNDALE** and **E. Krishna Iyer**, **araimandi**, solo · **KATHAK** **Uttar Pradesh - THE ONLY NORTHERN FORM**, from the **Kathakas**, **Hindu-Muslim synthesis**, **chakkars**, **gharanas** **LUCKNOW, JAIPUR, BANARAS** · **KATHAKALI** **Kerala** - male **dance-DRAMA**, **chutti** make-up · **KUCHIPUDI** **Andhra Pradesh** - **SIDDHENDRA YOGI**, **TARANGAM** on a brass plate · **ODISSI** **Odisha** - **TRIBHANGA** and **chauka**, **Jagannath**, **Mahari** and **Gotipua**, **Kelucharan Mohapatra** · **MANIPURI** **Manipur** - **Ras Leela**, gentle and rounded, restrained expression, **Pung Cholom** · **MOHINIYATTAM** **Kerala** - solo female, **lasya**, **Swathi Thirunal** · **SATTRIYA** **Assam** - **SRIMANTA SANKARADEVA**, performed in **sattras**, recognised 2000 · **KERALA HAS TWO; CHHAU** is the ninth candidate but is classed **FOLK/MARTIAL**

FOLK DANCES **Punjab** **Bhangra/Giddha** · **Gujarat** **Garba/Dandiya** · **Rajasthan** **Ghoomar/Kalbelia** · **Maharashtra** **Lavani/Tamasha** · **Assam** **Bihu** · **Odisha-Jharkhand-WB** **Chhau** · **Karnataka** **Yakshagana/Dollu Kunitha** · **Kerala** **Theyyam/Ottamthullal** · **Mizoram** **Cheraw** bamboo · **Tripura** **Hojagiri** · **J&K** **Rouf** · **UP** **Nautanki** · **Bihar** **Bidesia** · **AP** **Burra Katha** · **UNESCO ICH**: **Kutiyattam**, **Vedic chanting**, **Ramlila**, **Ramman**, **Chhau**, **Kalbelia**, **Mudiyettu**, **Ladakh Buddhist chanting**, **Sankirtana**, **Thatheras**, **Yoga**, **Kumbh Mela**, **Durga Puja Kolkata** and **Garba of Gujarat (2023) - the two most recent**

MUSIC - one root, two systems descends from the **SAMA VEDA** via the **Natyashastra** and the **SANGITA RATNAKARA** of **SARANGADEVA (13th c)**, which **PREDATES the split and is common ancestor to both** · division from the **13th-14th centuries** under **Persian influence** · both share **RAGA** (melodic framework) and **TALA** (rhythmic cycle) · **HINDUSTANI: improvisation**, extended **alap**, **10 THAATS (BHATKHANDE)**, forms **khayal** (dominant), **dhrupad** (older, austere), **thumri**, **tarana**, **dhamar**, **tappa** · instruments **sitar**, **sarod**, **sarangi**, **santoor**, **shehnai**, **tabla** · **gharanas** **Gwalior**, **Kirana**, **Jaipur-Atrauli**, **Agra** · **TANSEN** at **Akbar's court** · **CARNATIC: composition-centred**, the **kriti**, **72 MELAKARTAS (VENKATAMAKHIN)**, forms **varnam**, **kriti**, **ragam-tanam-pallavi** · instruments **veena**, **mridangam**, **violin**, **ghatam**, **nadaswaram** · **PURANDARA DASA** = "father of Carnatic music" but NOT of the Trinity · **THE TRINITY = TYAGARAJA, MUTHUSWAMI DIKSHITAR, SYAMA SASTRI**

INSTRUMENT CLASSES (Natyashastra, and the same basis as modern Hornbostel-Sachs) **TATA** strings - **veena**, **sitar**, **sarod**, **violin** · **SUSHIRA** wind - **flute**, **shehnai**, **nadaswaram** ·

AVANADDHA stretched membrane - **tabla**, **mridangam**, **pakhawaj** · **GHANA** solid, no membrane - **manjira**, **jhanj**, **ghatam**, **jaltarang** · **the discriminator: avanaddha has a skin, ghana does not**

PAINTING SCHOOLS MUGHAL - AKBAR founds the atelier, **Persian technique + Indian subjects**, **Hamzanama**, **Razmnama**, artists **Basawan** and **Daswanth** · **JAHANGIR THE PEAK - naturalism**, **portraits**, **birds and animals**, **USTAD MANSUR** and **Abul Hasan** · **SHAH JAHAN** formal and jewel-like · **AURANGZEB withdrew patronage and the dispersal of his artists enriched the Rajput and Deccani courts** · **RAJASTHANI/RAJPUT** - **Mewar**, **Bundi**, **Kota** (hunts), **KISHANGARH (BANI THANI)**, **Marwar**, **Jaipur**; themes **Krishna**, **Ragamala**, **Barahmasa** · **PAHARI - BASOHLI** bold with beetle-wing green, **KANGRA** lyrical, **Nainsukh's family**, **Gita Govinda**; **Guler**, **Chamba**, **Garhwal** · **DECCANI** - **Ahmadnagar**, **BIJAPUR**, **GOLCONDA**, **Hyderabad** · **COMPANY STYLE** Indo-European for British patrons · **TANJORE - GOLD FOIL over gesso**

FOLK PAINTING - the four to know cold **MADHUBANI = BIHAR** · **WARLI = MAHARASHTRA** · **PATTACHITRA = ODISHA** · **KALAMKARI = ANDHRA PRADESH** (**Srikalahasti** freehand, **Machilipatnam** block-printed) · then recognition only: **PHAD** **Rajasthan** · **GOND** **MP** · **KALIGHAT** **Bengal** · **CHERIYAL** **Telangana** · **SAURA** **Odisha** · **SOHRAI-KHOVAR** **Jharkhand** · **AIPAN** **Uttarakhand** · **PITHORA** **Gujarat** and **MP** · **THANGKA** **Ladakh** and **Sikkim**

8. Five-Minute Wall Sheet

FIVE-MINUTE WALL SHEET · CHAPTER F5.6 - TEXTS, SCIENCE AND THE PERFORMING ARTS

INDO-ARYAN north (Sanskrit, Prakrit, Pali) · DRAVIDIAN south = SEPARATE FAMILY, not from Sanskrit

SANSKRIT ritual · PRAKRIT Ashoka's edicts + Jain canon · PALI Buddhist Tripitaka

CLASSICAL LANGUAGES NOW ELEVEN (was six until October 2024)

TAMIL 2004 first · SANSKRIT 2005 · TELUGU + KANNADA 2008 · MALAYALAM 2013 · ODIA 2014

Added 2024: MARATHI, PALI, PRAKRIT, ASSAMESE, BENGALI · ODIA = first Indo-Aryan after Sanskrit

RAMAYANA-VALMIKI · MAHABHARATA-VYASA, longer, CONTAINS THE BHAGAVAD GITA

MANUSMRITI = Dharmashastra · PANINI Ashtadhyayi · PATANJALI Mahabhashya

DECIMAL PLACE VALUE + ZERO = the great Indian contribution, to Europe via the ARABS

BRAHMAGUPTA = rules for ZERO and NEGATIVES (Brahmasphutasiddhanta)

ARYABHATA = Aryabhata, EARTH ROTATES, ECLIPSES AS SHADOWS, pi 3.1416

SULBASUTRAS = altar geometry, Pythagorean relation as a CONSTRUCTION RULE

BHASKARA II = Siddhanta Shiromani · MADHAVA / KERALA SCHOOL = INFINITE SERIES

VARAHAMIHIRA = Brihatsamhita · JANTAR MANTAR = 18th c, SAWAI JAI SINGH II, NOT ancient

CHARAKA = MEDICINE (Kushana) · SUSRUTA = SURGERY + RHINOPLASTY

MEHRAULI PILLAR = PHOSPHATE layer · WOOTZ STEEL → Damascus blades

OVERSTATED: no ancient aircraft - the Vaimanika Shastra is 20th century

NALANDA = KUMARAGUPTA I, described by HIUEN TSANG, sacked c. 1200

VIKRAMASHILA + ODANTAPURI = PALA · VALABHI = 2nd Jain Council 512 CE ·

TAKSHASHILA earliest

NATYASHASTRA = BHARATA · NRITTA pure, NRITYA expressive, NATYA drama · NINE RASAS

EIGHT CLASSICAL DANCES (Chhau is the folk ninth)

BHARATANATYAM Tamil Nadu · oldest · RUKMINI DEVI ARUNDALE revived it

KATHAK Uttar Pradesh = THE ONLY NORTHERN FORM · gharanas LUCKNOW, JAIPUR, BANARAS

KATHAKALI Kerala = male DANCE-DRAMA, chutti make-up

KUCHIPUDI Andhra Pradesh = SIDDHENDRA YOGI, TARANGAM on a brass plate

ODISSI Odisha = TRIBHANGA posture · MANIPURI Manipur = Ras Leela, gentle

MOHINIYATTAM Kerala = solo female, lasya · KERALA HAS TWO FORMS

SATTTRIYA Assam = SANKARADEVA, satras, recognised 2000

Folk: Bhangra Punjab · Garba Gujarat · Ghoomar Rajasthan · Lavani Maharashtra · Bihu Assam

MUSIC from the SAMA VEDA · SANGITA RATNAKARA (Sarangadeva, 13th c) PREDATES the split

HINDUSTANI: improvisation · 10 THAATS (BHATKHANDE) · KHAYAL · sitar, tabla · TANSEN

CARNATIC: composition · 72 MELAKARTAS (VENKATAMAKHIN) · kriti · veena, mridangam

CARNATIC TRINITY = TYAGARAJA, MUTHUSWAMI DIKSHITAR, SYAMA SASTRI

PURANDARA DASA = "father of Carnatic music" but NOT in the Trinity

TATA strings · SUSHIRA wind · AVANADDHA has a SKIN (tabla, mridangam) · GHANA has none (manjira, ghatam)

MUGHAL: AKBAR founds · JAHANGIR THE PEAK, naturalism, USTAD MANSUR · AURANGZEB ends it

KISHANGARH = BANI THANI · BASOHLI bold · KANGRA lyrical · GOLCONDA = DECCANI ·

TANJORE = GOLD FOIL

MADHUBANI Bihar · WARLI Maharashtra · PATTACHITRA Odisha · KALAMKARI AP